



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
HALF YEARLY EXAMINATION

CLASS: X
SUBJECT: GEOGRAPHY (SET A)

FULL MARKS: 80
TIME: 2 HOURS

*Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the question paper.*

*The time given at the head of this Paper is the time allowed for writing the answers.
Part I is compulsory. All questions from Part I are to be attempted.
A total of five questions are to be attempted from Part II.
The marks intended for questions are given in brackets [].
This paper consists of six printed pages.*

PART I (30 marks)
(Attempt all questions from this Part.)

Question 1

Study the map extract G43S7 of the Survey of India and answer the following questions:

- (i) (a) Give the six figure grid reference of $\Delta 268$. [2]
(b) How is the drainage pattern in the grid square 3697 different from grid 3899?
- (ii) How is the settlement pattern of grid square 3994 different from grid square 3591? [2]
- (iii) (a) State the compass direction of Vadavas from Ratanpur and Nani Bhatamal. [2]
(b) Calculate the area between Eastings 33 to 38 and Northings 92 to 98 in km.
- (iv) (a) Name any two types of natural vegetation present in the map extract. [2]
(b) What type of rainfall is experienced in the given map area? Give two evidences to support your answer.
- (v) Mention two manmade and two natural features in the grid square 3592. [2]

Question 2

On the outline map of India provided:

[10]

- (i) Shade an area of laterite soil in the north-east.
- (ii) Mark and label the tributary of River Yamuna.
- (iii) Mark and name the highest peak of Himalayas in India.
- (iv) Shade and label Gulf of Khambhat.
- (v) Mark and label Malabar Coast.
- (vi) Mark and label Indian Standard Meridian.
- (vii) Shade and label Andaman Sea.
- (viii) Mark and name an onshore oil field.
- (ix) Draw arrows to show the direction of the winds that bring rainfall to Coromandel Coast in winter.
- (x) Mark and label Karakoram Pass.

Question 3

Choose the correct answers to the questions from the given options.

[10]

(Do not copy the question, write the correct answers only.)

- (i) In which of the following states is the Kalpakkam Nuclear Power Plant located?
(a) Gujarat (b) Tamil Nadu
(c) Kerala (d) Odisha
- (ii) Which of the following factors cause the rise in the temperature of mainland India in the month of March?
(a) The land breeze blows towards the sea.
(b) Northward migration of the sun.
(c) Southward migration of the sun from the Equator.
(d) The pressure belts shift towards the east.
- (iii) Which of the following is correct about the laterite soil?
(a) It has poor moisture retaining capacity.
(b) It is formed by the weathering of crystalline and metamorphic rock.
(c) It is rich in lime.
(d) Mainly suitable for agriculture.
- (iv) Manganese is used to make ____ tough and resistant to rusting.
(a) Copper (b) Bauxite (c) Bronze (d) Steel
- (v) ____ is the oldest steel plant in India.
(a) CISCO (b) TELCO (c) TISCO (d) DSP
- (vi) Tidal energy can be harnessed at _____.
(a) Gulf of Kutch (b) Sundarbans
(c) Andaman and Nicobar Islands (d) All of the above
- (vii) Which soil has calcareous concretions and it is coarse in texture.
(a) Laterite soil (b) Red soil
(c) Khadar soil (d) Bhangar soil
- (viii) Name the following non-conventional source of energy that is used to generate electricity from the picture given below?



- (a) Geothermal energy (b) Solar energy
- (c) Biogas energy (d) Hydel energy

- (ix) **Assertion (A):** Parts of Punjab receive rainfall in winter due to temperate cyclones originating from the Mediterranean sea.
Reason (R): Himalayas receive a considerable amount of snowfall due to these cyclonic depressions.
 (a) Both A and R are true and R is the correct explanation of A.
 (b) Both A and R are true and R is not the correct explanation of A.
 (c) A is true but R is false
 (d) A is false but R is true
- (x) If you have to transport coal over a long distance, which mode of transport will you use and why?
 (a) Air transport because it is the fastest.
 (b) Rail transport because it is convenient to transport heavy and bulky commodities.
 (c) Road transport as it is economical.
 (d) Water transport as it is fuel efficient.

PART – II (50 marks)

(Attempt any five questions from this part)

Question 4

The monsoon's arrival in South Bengal has been delayed, with the India Meteorological Department (IMD) ruling out its onset in the region for at least five more days as of early July 2024. This delay contrasts with its timely onset in Kerala and parts of Northeast India. The IMD indicates that while North Bengal will see heavy rain, South Bengal will experience only scattered thunderstorms in the coming days.

India Meteorological Department, July 2024

- (i) (a) What caused this delayed monsoon in South Bengal? [2]
 (b) State two features of South West monsoon.
- (ii) Mention the different sources of rain in Punjab and Tamil Nadu during the winter season. [2]
- (iii) Give the geographical reasons for each of the following: [3]
 (a) The monsoon rain is unevenly distributed in India.
 (b) South-West monsoon withdraw from India in October.
 (c) Kochi has a lower range of temperature than Agra.
- (iv) Study the climatic data given below and answer the following questions: [3]

MONTHS	J	F	M	A	M	J	J	A	S	O	N	D
Temp in °C	14.7	16.6	21.6	25.5	33.2	39.5	30.8	29.8	29.2	25.5	19.6	15.2
RF in CM	25.3	16.9	1.5	0.9	1.5	7.5	17.8	18.5	12.5	1	0.2	24.9

- (a) Calculate the annual range of temperature.
 (b) Name the wind that brings rain to the station.
 (c) Suggest a probable name for this station.

Question 5

- (i) In the region of Nagpur, known for its rich, dark soil, farmers thrived on the fertile land. Among them was Arjun, who inherited a small plot renowned for its productive sugarcane and cotton crops. [2]
- (a) Identify the soil that is being discussed in the above passage.
- (b) Give two characteristic features of the above mentioned soil found most suitable for growing cotton and sugarcane.
- (ii) Define the following: [2]
- (a) Pedogenesis
- (b) Khadar soil
- (iii) With reference to Alluvial Soil, answer the following: [3]
- (a) How is it formed?
- (b) Write two characteristics of deltaic alluvium .
- (c) Name any two areas where it is found.
- (iv) Give the geographical reasons for each of the following: [3]
- (a) Alluvial soil varies in texture.
- (b) Man is largely responsible for soil erosion.
- (c) Black soil does not get leached.

Question 6

- (i) Mahesh and his friend Ravi decided to explore the majestic mountains near their town. After a strenuous hike, they reached a dense mountain forest. The air was cool and fresh, filled with the scent of pine and cedar. [2]
- (a) Which category of forests do pine and cedar belong to?
- (b) Mention the climatic condition that favours the growth of this kind of vegetation.
- (ii) (a) Why are the forests grown around the cities having the Iron and steel industries? [2]
- (b) Why do tropical evergreen forests appear dark inside the forests?
- (iii) What is Agroforestry? Write its two objectives. [3]
- (iv) (a) Name a forest conservation method that involves the local community. [3]
- (b) Why do the tropical desert forests have long roots?
- (c) Name the process in which large scale afforestation is done in areas unfit for agriculture.

Question 7

- (i) How are tube wells beneficial to farmers in North India? [2]
- (ii) Why are inundation canals converted to perennial canals? Give two reasons. [2]
- (iii) (a) State two reasons why irrigation is important in India. [3]
- (b) Name a state where tank irrigation is popular.
- (iv) Which irrigation method is known as overhead irrigation? Mention two advantages of it. [3]

Question 8

- (i) India is making significant strides in solar energy as part of its renewable energy strategy. The government aims to achieve 100 GW of solar capacity by 2022, positioning India as a global leader in solar power. This ambitious target is supported by initiatives such as

large-scale solar farms, rooftop solar installations and innovative applications like solar pumps for irrigation and solar street lighting. [2]

(a) How is India in an advantageous position to harness solar energy? [2]

(b) Name the first village in Andhra Pradesh which is fully electrified by solar energy. [2]

(ii) With reference to Bhakra Nangal project answer the following questions: [2]

(a) On which river it is based?

(b) Name any two Indian states benefited by this project.

(iii) (a) Name the leading producer of manganese in India. [3]

(b) Which variety of coal is popular for domestic use? Give a reason for your answer.

(c) Petroleum is referred to as liquid gold- justify the statement.

(iv) [3]



(a) Identify the type of energy being harnessed from the picture given above.

(b) Mention why this form of energy is preferred over nonrenewable energy.

(c) Write any one advantage of this type of energy.

Question 9

(i) What are footloose industries? Give one example . [2]

(ii) Mention two factors that have encouraged the growth of the IT industry in Bengaluru. [2]

(iii) (a) Name an Iron and Steel industry set up in Odisha with the help of a famous German firm. [3]

(b) From where does this industry get its iron ore and manganese?

(iv) Give the geographical reason for each of the following: [3]

(a) Products made from petrochemicals are growing in popularity.

(b) The electronic industry has made an impact on both entertainment and education.

(c) The iron and steel industry constitutes the backbone of the modern industrial economy.

Question 10

(i) State two advantages of inland waterways. [2]

(ii) Give two benefits of the Golden Quadrilateral project. [2]

(iii) (a) Which is the oldest port in India? [3]

(b) Name the organization responsible for the development, maintaining and management of highways in India.

(c) Name the port located about 50 km SouthWest of Kolkata.

- (iv) Give the geographical reason for each of the following: [3]
- (a) Railways are not common in North-East India.
 - (b) Airways are used extensively despite being an expensive mode of transportation.
 - (c) Roadways are the best for the distribution of milk and fresh flowers.

Question 11

- (i) The Coromandel Coast has more rainy months but less rainfall than the Konkan Coast. Why? [2]
- (ii) Why is well irrigation still a popular means of irrigation? Give two reasons to support your answer. [2]
- (iii) Name one centre in each of the following states where iron ore is mined: [3]
- (a) Odisha
 - (b) Jharkhand
 - (c) Karnataka
- (iv) (a) Name the raw materials required by the petrochemical industry. [3]
- (b) Why is a petrochemical industry usually located near an oil refinery?

Question 12

- (i) Differentiate between transported soil and in situ soil. [2]
- (ii) Give two ways in which forests are important. [2]
- (iii) State three advantages of roadways over railways. [3]
- (iv) Give the geographical reason for each of the following: [3]
- (a) Railway is an important means of transport compared to airways.
 - (b) The forest area has greatly depleted in India.
 - (c) Laterite soil is not suitable for cultivation.